

INTRODUCTION

Background

- Comprehensive understanding of atmospheric corrosion allows for accurate estimation of corrosion over time, promoting informed design choices and the implementation of effective anti-corrosion protection strategies.
- Despite advancements, there exists a knowledge gap in the corrosivity of metals in subarctic regions, such as Alaska, where the complexity of corrosion increases causing the evaluation of atmospheric corrosivity to be difficult.
- One primary factor affecting the rate of atmospheric corrosion is the alternating wet and dry cycles, which involves the time of wetness (TOW) and the formation of thin electrolyte layers.

Objectives

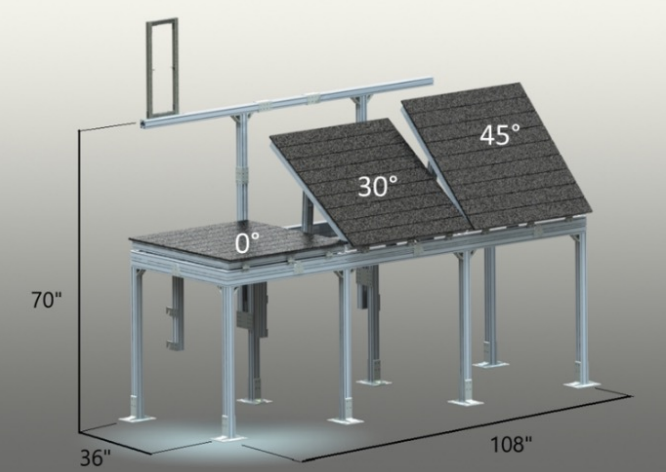
- To assemble and deploy multi-angle corrosion test racks with auxiliary weather stations at four test sites across Alaska, USA.
- To monitor and measure the air temperature, relative humidity, aerosol chloride and sulfate deposition, TOW at exposure angles of 0°, 30°, and 45°, and rainfall at the test sites across 6-month, 12-month, and 24-month exposure periods.
- To expose five different metal alloys to atmospheric conditions in Alaska at 0°, 30°, and 45° exposure angles.*
- To correlate the corrosion rate of metal alloys to exposure conditions and weather parameters to determine the corrosivity category and compare to recognized standards.
- To develop equations to more accurately model and estimate the corrosion rates and evaluate the corrosivity of the atmosphere in Alaska.

*Only 1008 Carbon Steel will be discussed in this poster

EXPERIMENTAL METHOD

Adjustable Atmospheric Test Rack Design

Measured Weather Parameters



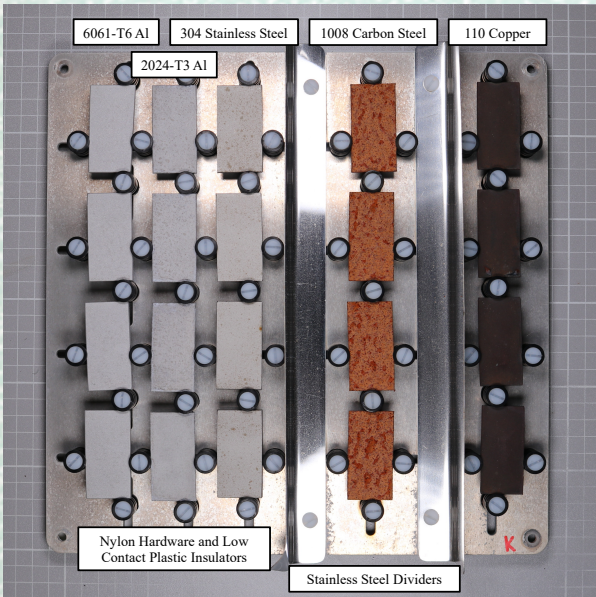
Weather Parameters Measured	
Air Temperature [°C]	
Relative Humidity [%]	
Aerosol Chlorides & Sulfates [mg/(m ² ×day)]	
Time of Wetness [mV]	
Rainfall [mm]	

Test Rack, Weather Station, and Metal Alloys

- Test rack design guided by ASTM G50 *Standard Practice for Conducting Atmospheric Corrosion Tests on Metals* [1] and ISO 8565 *Metals and Alloys – Atmospheric Corrosion Testing – General Requirements* [2].
- Test rack design built with adjustable aluminum rails that allow for exposure of metals at exposure angles of 0°, 30°, and 45°.
- Allows for support of a full weather monitoring system to remove reliance on the nearest meteorological observatories, increasing the accuracy of weather parameters related to coupon exposure. Chloride candles were prepared and used according to ASTM G140 *Standard Test Method for Determining Atmospheric Chloride Deposition Rate by Wet Candle Method* [3].
- 1" × 2" Coupon preparation consisted of sandblasting, cleaning with compressed air, pin stamping, degreasing, cleaning with acetone, and recording pre-exposure mass.
- Coupons were mounted while avoiding alteration, thus avoiding variability in corrosion. Dividers were also used to prevent cross-contamination of corrosion products.

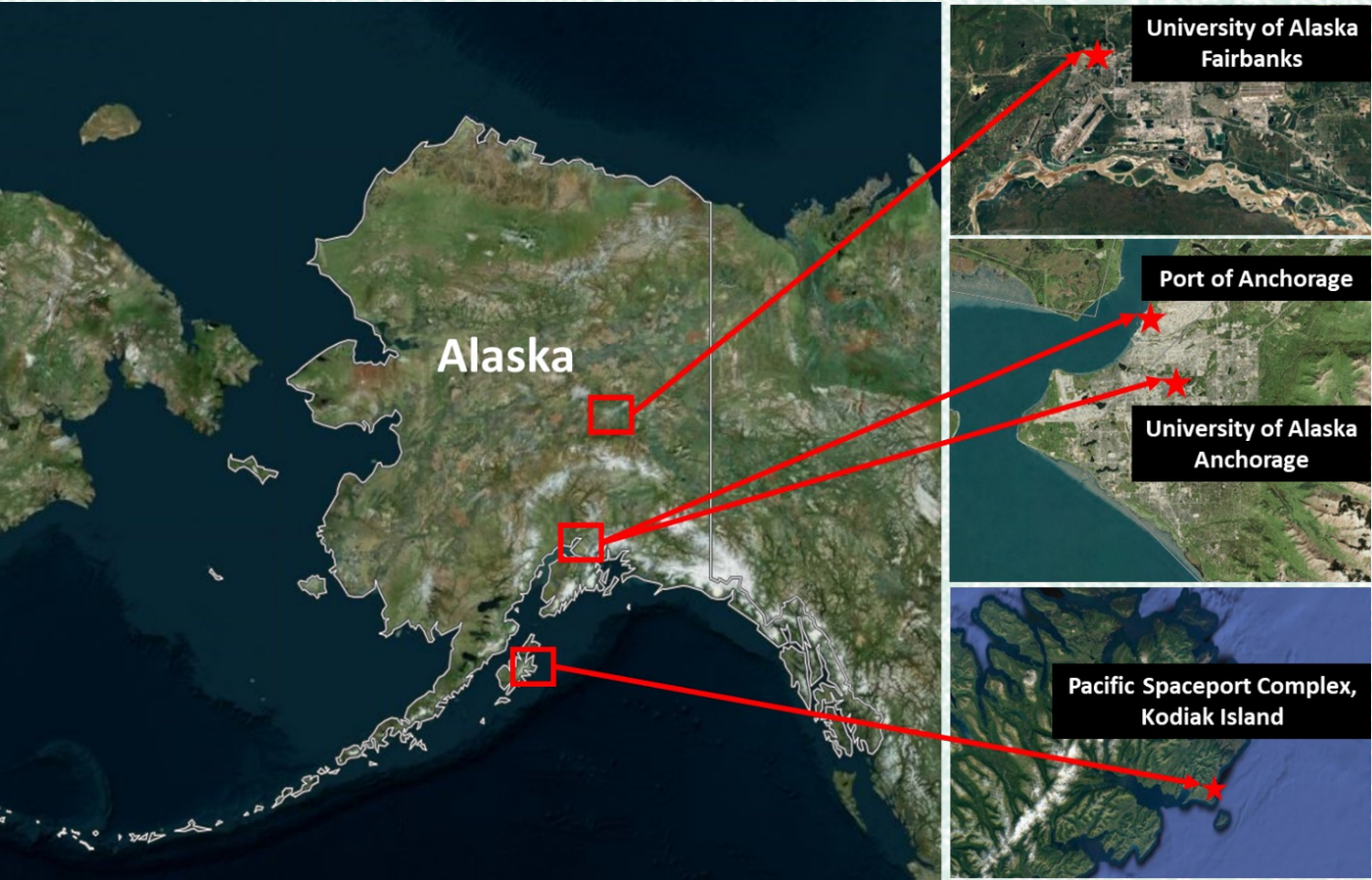
Mounted Metal Alloys with Hardware

Five Metal Alloys Studied



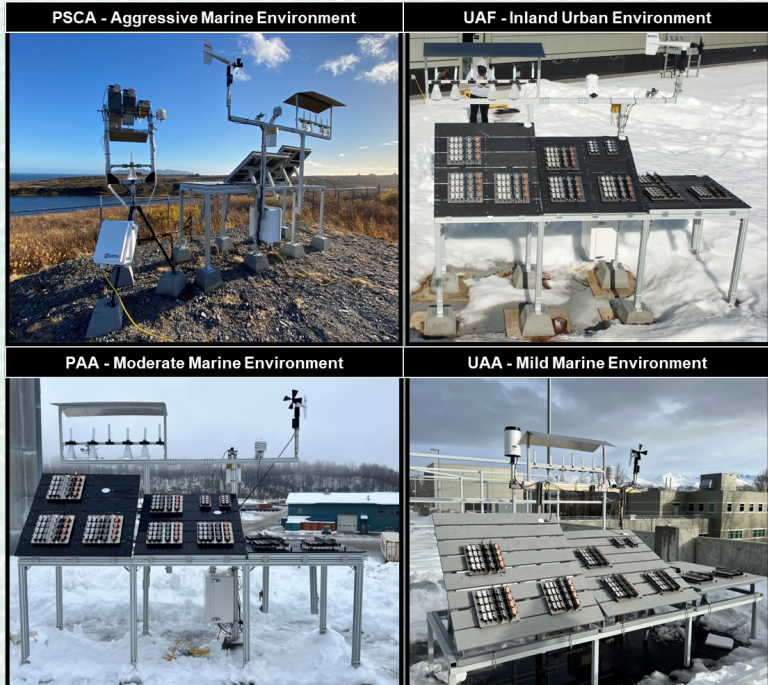
Metal Alloys
6061-T6 Aluminum Alloy UNS G96061
2024-T3 Aluminum Alloy UNS A92024
1008 Carbon Steel UNS G10080
304 Stainless Steel UNS G30400
110 Copper UNS C11000

Map of Alaska with Test Site Location



Test Site Location and Environment

Location	Coordinates	Environment
Pacific Spaceport Complex – Alaska [Kodiak, AK]	57.453°N, 152.346°W	Aggressive Marine Environment
Port of Alaska [Anchorage, AK]	61.239°N, 149.888°W	Moderate Marine Environment
University of Alaska Fairbanks [Fairbanks, AK]	64.855°N, 147.817°W	Far-inland Urban Environment
University of Alaska Anchorage [Anchorage, AK]	61.192°N, 149.822°W	Inland Mild-Marine Environment



Coupon Exposure Periods

Exposure Periods	
6 Month Set 1	Feb 2022–Aug 2022
6 Month Set 2	Aug 2022–Feb 2022
12 Month Set 1	Feb 2022–Feb 2023
12 Month Set 2	Feb 2023–Jan/Feb 2024
24 Month Set 1	Feb 2022–Jan/Feb 2024

Test Sites and Exposure Periods

- 4 strategic locations were selected as coupon exposure test sites, with test site selection and characterization influenced by ASTM G92 *Standard Practice for Characterization of Atmospheric Test Sites* [4].
- Test sites, spread around Alaska, were chosen to obtain a set of meteorological profiles that represent the diverse atmosphere and environment of Alaska along with their significance at a national and state level.
- 5 Exposure Periods were chosen to study the effects of corrosion over time.
- Each test site contained four coupons of each metal alloy for each exposure period at each exposure angle.

Determining Corrosion Rate

- Post exposure processing consisted of chemically cleaning according to ASTM G1 *Standard Practice for Preparing, Cleaning, and Evaluating Corrosion Test Specimens* [5].
- Each coupon mass was taken between each cleaning to ensure corrosion product was being removed while a stable mass was being approached.
- A pristine reference coupon was included in each respective batch of cleaning to verify the cleaning procedure was not significantly attacking any base metal.
- Cleaned coupons were weighed and the corrosion rate, in grams per meter squared per day (gmd), was calculated using equation 1 where M represents the respective coupon weight, SA represents the total surface area, and D represents the exposure period in days.

$$\text{Corrosion Rate} = \frac{M_{\text{initial}} - M_{\text{final}}}{SA \cdot D} \quad (1)$$

RESULTS & DISCUSSION

Summary of Weather Data at each Test Site over the 12 Month Set 1 and Set 2 Exposure Periods

Location	Set	Exposure Period [Days]	Average Air Temperature [°C]	Average Relative Humidity [%]	Chloride Deposition [mg/(m ² ×day)]	Sulfate Deposition [mg/(m ² ×day)]	Time of Wetness [%]		
							0°	30°	45°
Pacific Spaceport Complex – Alaska [Kodiak, AK]	12M S1	365	6.26	85.01	47.73	14.99	54.20	48.20	41.41
	12M S2	357	5.26	81.11	42.78	12.56	45.12	37.74	30.74
	12M S1	370	4.22	76.93	11.33	15.67	28.09	29.75	21.93
Port of Alaska [Anchorage, AK]	12M S2	337	3.71	76.82	10.66	12.95	40.08	35.76	25.98
	12M S1	376	-0.07	66.83	9.26	22.89	28.09	29.75	21.93
University of Alaska Fairbanks [Fairbanks, AK]	12M S2	357	-0.25	67.58	10.76	19.14	40.08	35.76	25.98
	12M S1	367	4.05	74.16	8.85	15.09	31.96	43.52	18.29
University of Alaska Anchorage [Anchorage, AK]	12M S2	335	3.36	77.31	10.49	9.82	50.02	50.78	40.98

Weather Data

- Temperature readings present the relative frigidity in Alaska where when aerosol particles are present, the freezing point of a liquid layer on a metal's surface is reduced well below freezing.
- Relative humidity is an important factor when coupled with the presence of hygroscopic contaminants, fostering an environment for corrosion on metal surfaces.
- Chloride ions and sulfates are the two leading contaminants that impact corrosion. The PSCA test site resulted with four to six times that of the PAA, UAF, and UAA test sites due to its proximity to the ocean.
- Complexity in corrosion can be attributed to rainfall, as its ability to cleanse dry deposits not only depends on precipitation patters but also on the characteristics of the corrosion products.
- ISO 9223 *Corrosion of Metals and Alloys – Corrosivity of Atmospheres – Classification, determination and estimation* [6], indicates TOW can be inferred from temperature and relative humidity.
- The PSCA test site registered the highest TOW, corresponding with its highest average air temperature and relative humidity.

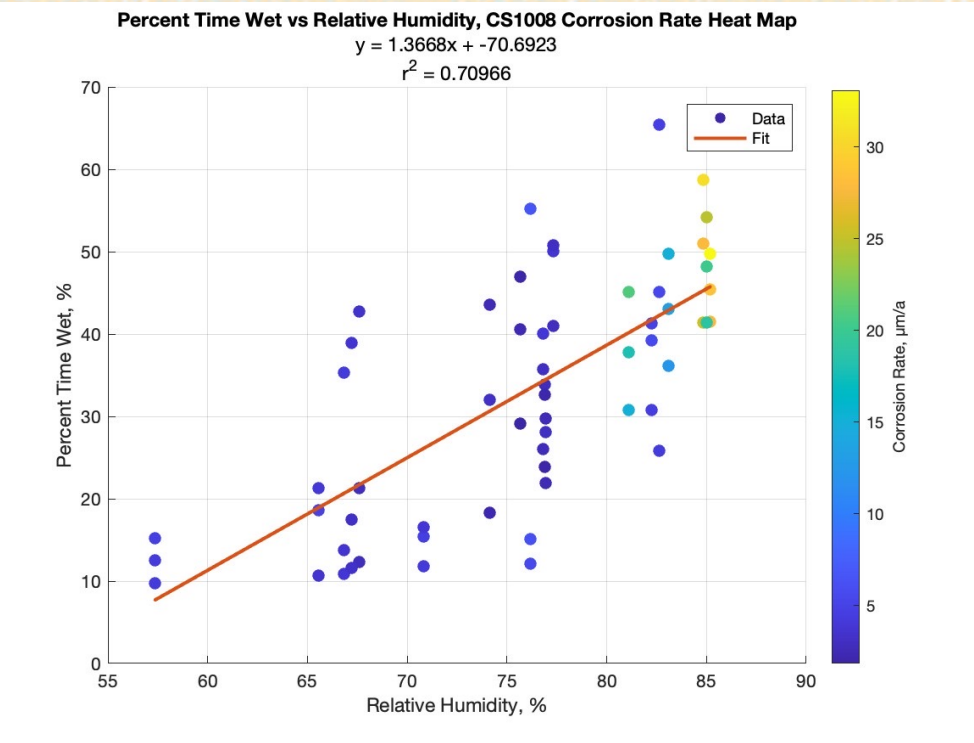
Corrosion Rates

- Carbon steel coupons exposed at the PSCA test site exhibited corrosion rates four times greater than those exposed at the PAA, UAF, and UAA test sites. This can be attributed to the PSCA test site recording four times the chloride deposition rate as well as the coupons having spent 18% more time wet than the others.
- The TOW was found to inversely correlate with the exposure angle. The coupons deployed at a 0° exposure angle resulted with a higher TOW than those at a 30° and 45° angle.
- Longer exposure periods correlated with lower corrosion rates, highlighting the effect of protective corrosion product layers.

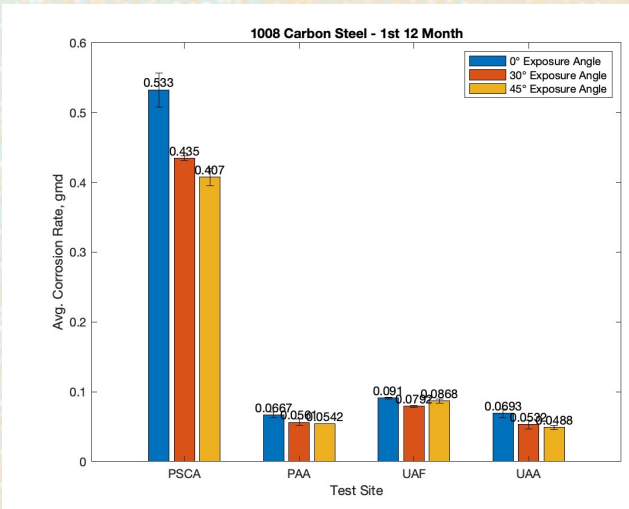
Corrosion Estimation and Corrosivity Category

- ISO 9223 [6] states that the corrosivity of an atmosphere shall be classified through determination of corrosivity category either based on experimental exposure of specimens or by estimation based on weather parameters and use of a dose-response function, seen in equation 2 for carbon steel, with C1 (very low) to CX (extreme).
- Compared to the corrosion rates determined through coupon exposure, the dose-response function estimates the corrosion rates to be two 2-3 times greater for the PSCA and UAF test sites and 6-7 times greater for the PAA and UAA test sites.
- To improve the accuracy of corrosion rate estimation, nonlinear regression analysis with parameter optimization was used to fit a second-order quadratic model with interaction terms. This model the represents experimental corrosion rates as a function of air temperature (T), sulfate deposition (Pd), chloride deposition (Sd), percent time wet (PTW), and exposure duration in years (where 1 year = 365 days), as seen in equation 3 for carbon steel.
- This model substitutes relative humidity with percent time wet, providing a more accurate representation of metal moisture exposure. Additionally, a time variable was added to account for the effects of protective corrosion product layers.

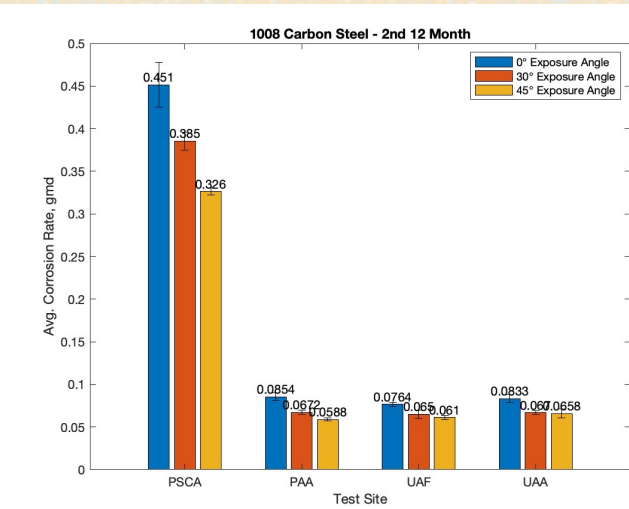
PTW/RH Correlation, CS1008 Corrosion Rate



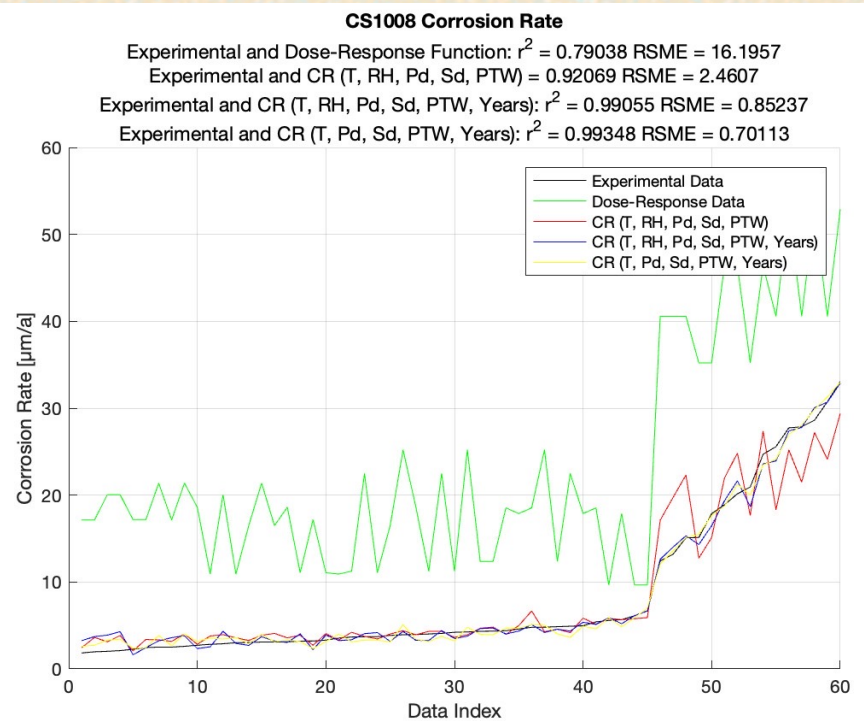
CS1008 12 Month Set 1 Corrosion Rate



CS1008 12 Month Set 2 Corrosion Rate



Comparison of Corrosion Rate Estimation Methods



1008 Carbon Steel Corrosion Rate and Corrosivity Categorization – 12 Month Set 1 and Set 2

1008 Carbon Steel Corrosion Rate – 12 month Set 1 & 2									
Location	Set	Exposure Angle [°]	Experimental		Dose-Response		Optimized Equation		CC
			Corrosion Rate [g/(m ² ×year)]	CC	Corrosion Rate [g/(m ² ×year)]	CC	Corrosion Rate [g/(m ² ×year)]	CC	
Pacific Spaceport Complex – Alaska [Kodiak, AK]	12M S1	0	194.42	C2	365.29	C3	186.08	C2	
	12M S2	0	164.76		277.19		157.09		
	12M S1	30	158.63		365.29		168.46		
	12M S2	30	140.69		277.19		138.69		
	12M S1	45	148.72		365.29		148.94		
	12M S2	45	119.04		277.19		121.71		
Port of Alaska [Anchorage, AK]	12M S1	0	24.35	C2	168.27	C2	31.31	C2	
	12M S2	0	31.18		146.43		27.12		
	12M S1	30	20.48		168.27		31.53		
	12M S2	30	24.52		146.43		26.50		
	12M S1	45	19.77		168.27		30.71		
	12M S2	45	21.46		146.43		25.77		
University of Alaska Fairbanks [Fairbanks, AK]	12M S1	0	33.21	C2	88.52	C2	24.46	C2	
	12M S2	0	27.90		85.94		33.48		
	12M S1	30	28.91		88.52		24.32		
	12M S2	30	23.73		85.94		30.47		
	12M S1	45	31.69		88.52		24.62		
	12M S2	45	22.28		85.94		30.47		
University of Alaska Anchorage [Anchorage, AK]	12M S1	0	25.28	C2	135.29	C2	19.68	C2	
	12M S2	0	30.40		129.85		25.99		
	12M S1	30	19.41		135.29		20.04		
	12M S2	30	24.44		129.85		26.15		
	12M S1	45	17.82		135.29		20.82		
	12M S2	45	24.02		129.85		24.50		

$$r_{corr}(T, RH, Pd, Sd) = 1.77 \times Pd^{0.52} \times e^{0.02 \times RH + f_{st}} + 0.102 \times Sd^{0.62} \times e^{0.033 \times RH + 0.040 \times T} \quad (2)$$
$$f_{st} = 0.150 \times (T - 10) \text{ when } T \leq 10^\circ\text{C}; \text{ otherwise } -0.054 \times (T - 10)$$

$$r_{corr}(T, Pd, Sd, PTW, \text{Years}) = 1.55 + (-0.113 - 0.416 \times T - 0.354 \times Pd + 0.598 \times Sd - 0.142 \times PTW + 0.044 \times T^2 + 0.009 \times Pd^2 - 0.005 \times Sd^2 + 0.0004 \times PTW^2 + 0.040 \times (T \times Pd) - 0.048 \times (T \times Sd) + 0.001 \times (T \times PTW) + 0.0003 \times (Pd \times Sd) + 0.002 \times (Pd \times PTW) + 0.009 \times (Sd \times PTW)) \times \text{Years}^{-0.566} \quad (3)$$

- Significant differences resulted between the two corrosion rate determination methods in ISO 9223 [6]. Compared to experimental results, the corrosivity category for the PAA, UAF, and UAA test sites remain in the C2 corrosivity category while the PSCA test site increases to the C3 corrosivity category using the dose-response function.
- The optimized equation model was able to provide more accurate corrosion rate estimation and match the experimental categorization for atmospheric corrosivity in Alaska.
- Similar results were concluded with the additional metals analyzed in this study.

References

- ASTM G50, "Standard Practice for Conducting Atmospheric Corrosion Tests on Metals," ASTM International, 2020
- ISO 8565, "Metals and Alloys. Atmospheric Corrosion Testing. General Requirements," BSI Standards Limited, 2011.
- ASTM G140, "Standard Test Method for Determining Atmospheric Chloride Deposition Rate by Wet Candle Method," ASTM International, 2019.
- ASTM G92, "Standard Practice for Characterization of Atmospheric Test Sites," ASTM International, 2020.
- ASTM G1, "Standard Practice for Preparing, Cleaning, and Evaluating Corrosion Test Specimens," ASTM International, 2017.
- ISO 9223, "Corrosion of Metals and Alloys—Corrosivity of Atmospheres—Classification, determination and estimation," 2012.

Future Work

- Future work will involve coupon deployment in additional test sites throughout Alaska to further increase corrosion rate estimation accuracy and categorization of atmospheric corrosivity to represent the subarctic and arctic regions in Alaska.
- Future work will involve additional model equation testing to refine the regression analysis through parameter optimization and determine the limits of the model's applicability.

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Potential Future Test Sites

